The initiating codon is AUG and the initiating amino acid is formyl methionine.

- Elongation is facilitated by factors that facilitate the binding of aminoacyl tRNA to the A site on the ribosome as well as the movement of the ribosome along the mRNA.
- Peptide bond formation is catalyzed by peptidyl transferase and after peptide bond formation, it moves forward along the mRNA in 5' to 3' direction to the next codon.
- Termination begins when one of the stop codons moves into the A site. The recognition of these codons by release factors releases the newly formed protein from the m-RNA-ribosome complex.
- Post-translational modifications like removal of amino acids, phosphorylation, glycosylation, hydroxylation and protein folding also occur after translation in eukaryotes.
- Recombinant DNA technology: This technology, is also known as DNA cloning,allows insertion of a DNA fragment of interest, such as a new gene, into an organism. The simplest strategy is introducing a new gene into bacteria through plasmid vectors. An outline of gene cloning using plasmid vectors is given.
- PCR amplification of DNA, sequencing of DNA using Sanger's dideoxy chain termination method and the latest Next Generation Sequencing (NGS) methods are explained along with their applications.

EVALUATION

I Fill in the Blanks

1. The information in the DNA can be copied to RNA by a process called _____
2. Name the three different models for DNA replication.
3. A short piece of RNA produced by primase enzyme is called as _____
4. The fragments of DNA in the lagging strand during DNA replication are called _____
5. _____ is an enzyme that strips away the primer(s).

II State True or False

6. In Prokaryotes, DNA replication occurs inside the nucleus.
7. Replication is very slow in Eukaryotes.
8. The eukaryotic mRNAs are mostly monocistronic.

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